Kaushal Francis J

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PROFESSIONAL SUMMARY

Agentic Fullstack Engineer with production experience designing **LLM orchestration workflows**, **RAG pipelines**, and **event-driven backend systems**. Proven ability to architect **tool-calling ecosystems** and robust applications, improving RAG accuracy by 40% in enterprise applications. Strong foundation in Python, TypeScript, and vector databases, applying systems thinking to build resilient customer-facing AI agents.

CORE COMPETENCIES

Agent Runtimes | Multi-provider LLM Orchestration | Retrieval & Ranking | Event-Driven Workflows | Vector Databases | Observability

WORK EXPERIENCE

Payday Suite - 4i apps solutions

April 2025 - Present

Software Engineer to Senior Associate (Promoted 2026)

Chennai, Tamil Nadu

- Improved **RAG chatbot response accuracy by 40%** in production by incorporating contextual metadata (module, title, content type) into embeddings and refining similarity scoring algorithms.
- Engineered an enterprise **document ingestion and chunking pipeline** processing structured documents into PgVector using AWS Bedrock Titan embeddings (1024-dim), featuring metadata-aware indexing for high-precision retrieval.
- Built a **custom LLM-as-a-judge evaluation harness** scoring Faithfulness and Answer Relevancy (1-5 rating in structured JSON) via Groq API, establishing automated validation for AI outputs without RAGAS overhead.
- Designed and integrated an **event-driven processing pipeline** using AWS SQS and Lambda, enabling asynchronous execution and increasing system reliability for enterprise processing.
- Developed a **configurable dynamic form engine and API layer** for nested DB schemas without redeployment, contributing tenant-based feature access control within the AI service layer.
- Owned **fullstack module architecture** end-to-end using Next.js, Node.js, Express.js, and MySQL, leading REST API design, database schema migrations, and production releases.

PROJECTS

NVIDIA NIM Desktop Coding Agent

Developed a desktop AI agent featuring full-duplex asynchronous token streaming and local workspace management. Engineered an LLM provider interface using Nvidia NIM APIs and integrated the **Model Context Protocol (MCP)** to dynamically expand agent capabilities with external tools. Architected a fault-tolerant streaming structure buffering non-deterministic JSON chunks, ensuring zero-latency UI updates during agentic reasoning.

Python, Flet, AsyncOpenAI, Model Context Protocol (MCP)

RAG Pipeline with Custom LLM Evaluation

Architected an end-to-end **RAG pipeline** - PDF ingestion, document-structure-aware chunking (fixed-size, recursive, semantic), FAISS vector retrieval, and LLM response generation via Groq API. Implemented **custom LLM-as-a-judge evaluation engine** scoring Faithfulness and Answer Relevancy in JSON format, analyzing token distribution and accuracy trade-offs across chunking strategies.

Python, Groq API, FAISS, Sentence Transformers

Production Habit Formation API (HabitFlow)

Built a production-oriented habit tracking API with timezone-aware streak calculation, configurable grace periods, and weekly/daily recurrence logic. Implemented layered architecture (controller to service to repository) with JWT session management, in-memory TTL cache, and Redis streak caching with graceful DB fallback. Enforced enterprise security best practices - bcrypt password hashing with full history reuse prevention, composite rate limiting, and Zod validation across all endpoints.

Node.js, TypeScript, Express.js, MySQL, Redis

EDUCATION

B. Tech, Computer Science (Cyber Physical Systems) Vellore Institute of Technology, Chennai 2021-2025

SKILLS

AI & GenAI: Agent Frameworks, RAG Systems, Contextual Embeddings, Vector Databases (PgVector, FAISS), LLM APIs (Groq, Gemini, Bedrock Titan), Custom LLM Evaluation, Tool-Calling

Backend (Python): Python, FastAPI, Sentence Transformers, Async I/O

Backend (TypeScript): JavaScript (ES6+), TypeScript, Node.js, Express.js **Databases & Cache:** PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: AWS (Lambda, SQS), Event-Driven Architecture, Microservices, Docker, Git